

Compatible orderings of binary vector spaces

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Abstract

A total order on a finite vector space is called compatible when the unique order-preserving bijection between any two subspaces of the same dimension is linear. Cameron asked for the number of compatible orders on $V(n, q)$ and observed that the answer was known only in dimensions one and two. We solve the problem over the binary field. For every $n \geq 4$, compatible total orders on $\mathbb{F}_2^n$ form exactly eight orbits under $\text{GL}(n, 2)$, and therefore their number is

$$8|\text{GL}(n, 2)| = 8 \prod_{i=0}^{n-1} (2^n - 2^i).$$

Four orbits have zero as their least element. They are represented by two explicit flag orders and by reversing the nonzero parts of those orders; the remaining four are obtained by reversing the full order. The proof has one finite computer-assisted base lemma: an exhaustive, symmetry-broken classification in dimension four, checked by two implementations using different invariants of ordered Fano planes. A local line condition then forces an initial complete flag, and a uniqueness lemma shows that each of the two relevant three-dimensional types has at most one lift in every higher dimension. The counts in dimensions 1, 2, 3 are respectively 2, 12, 10080.

1 Introduction

Let $V = V(n, q)$ be an n -dimensional vector space over $\mathbb{F}_q$. A total order $\prec$ on V is *compatible* if, whenever $U, W \leq V$ have the same dimension, the unique bijection from U to W that preserves their induced orders is a linear isomorphism. Cameron introduced this condition while studying monotone elements in inverse semigroups of partial linear isomorphisms [1]. Lexicographic orders give compatible orders over every finite field. Cameron also determined the possible position of zero, classified dimensions one and two, and noted that over $\mathbb{F}_2$ every order with zero extremal is compatible in dimension three. The counting problem was restated as Problem 30.7 of the BCC30 problem collection, where the answer was recorded as known only for $n = 1, 2$ [2].

We give a complete answer for $q = 2$. The binary case is rigid for two related reasons. First, the one-dimensional condition forces zero to be the least or greatest vector. Second, an ordered three-dimensional subspace is an ordered Fano plane, so its linear type can be encoded by the positions of its seven lines. An exhaustive four-dimensional base classification leaves four types with zero least. Two of them satisfy a local condition saying that the first three nonzero vectors in every three-space form a line; the other two are obtained by reversing all nonzero vectors. The local condition forces the initial segments of sizes $2^r - 1$ to be subspaces. Once this flag is present, the order outside its initial hyperplane is uniquely determined by the common ordered Fano type.

The computer calculation is confined to dimension four. Searches in dimensions five through seven were useful during discovery, but they are not used to infer the general theorem. The general step is the local-to-global flag argument and the unique-lift lemma proved below.

2 Compatible orders and two explicit families

Write vectors in coordinates $(x_1, \dots, x_n)$ and identify a binary vector with the integer $\sum_{i=1}^n 2^{i-1} x_i$. For $x \neq 0$, let

$$h(x) = \max\{i : x_i = 1\}.$$

Define two total orders with zero first. In L_n , nonzero vectors are ordered first by increasing $h(x)$ and then, within a fixed layer, by increasing key

$$\lambda_h(x) = \sum_{i < h} 2^{i-1} x_i.$$

In R_n , the layers are again ordered by increasing $h(x)$, but $\lambda_h(x)$ is decreasing inside each layer. Let ρ keep zero first and reverse the order of all nonzero vectors. Let ι reverse the full total order.

For example, the nonzero parts in dimension four begin

$$L_4 = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

and

$$R_4 = (1, 3, 2, 7, 6, 5, 4, 15, 14, 13, 12, 11, 10, 9, 8).$$

After the standard $\text{GL}(4, 2)$ normalization by the first four independent vectors, the second sequence becomes

$$(1, 2, 3, 4, 5, 7, 6, 8, 9, 11, 10, 14, 15, 13, 12).$$

Proposition 2.1. *The orders $L_n, R_n, \rho L_n, \rho R_n$ and their full reversals are compatible for every n .*

Proof. Let $W \leq \mathbb{F}_2^n$ have dimension d . Choose its reduced echelon basis $w_1, \dots, w_d$ using the highest nonzero coordinate as pivot, with the pivot positions strictly increasing. The coordinate map

$$\phi_W : \sum_{i=1}^d c_i w_i \mapsto (c_1, \dots, c_d)$$

is linear. The highest nonzero coordinate of a linear combination is the pivot belonging to the largest index i with $c_i = 1$. If two combinations have the same largest active pivot, their highest differing lower coordinate is the pivot belonging to the largest lower coefficient on which they differ. It follows that ϕ_W carries the order induced by L_n to L_d and the order induced by R_n to R_d .

For two d -subspaces U, W , the order-preserving bijection is therefore $\phi_W^{-1} \phi_U$, hence is linear. Reversing the nonzero parts of the induced orders on both subspaces leaves the same rank-matching bijection, and reversing both full orders does the same. Thus all eight displayed orders are compatible. $\square$

3 The four-dimensional base classification

The next lemma is the only computer-assisted component.

Lemma 3.1 (Four-dimensional base). *With zero fixed as the least element, compatible total orders on $\mathbb{F}_2^4$ form exactly four $\text{GL}(4, 2)$ -orbits, represented by*

$$L_4, \quad R_4, \quad \rho L_4, \quad \rho R_4.$$

Certificate and proof method. An ordering of the seven nonzero points of $\mathbb{F}_2^3$ is classified up to $\text{GL}(3, 2)$ by the set of position triples whose vectors sum to zero. This line-position code is complete because a bijection of the Fano plane that preserves its lines is linear. The $7!$ orders split into $7!/168 = 30$ types.

For an order of the 15 nonzero vectors of $\mathbb{F}_2^4$, global linear symmetry is removed by sending the first four independent vectors to 1, 2, 4, 8. For each of the 30 possible ordered Fano types, a prefix search adds vectors one at a time. Every hyperplane prefix must remain extendible to that same Fano type. This is exhaustive after the stated symmetry breaking. It gives the four normalized sequences

$$\begin{array}{cccccccccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ 1 & 2 & 3 & 4 & 5 & 7 & 6 & 8 & 9 & 11 & 10 & 14 & 15 & 13 & 12 \\ 1 & 2 & 4 & 7 & 8 & 11 & 13 & 14 & 9 & 10 & 12 & 15 & 5 & 6 & 3 \\ 1 & 2 & 4 & 7 & 8 & 11 & 13 & 14 & 15 & 12 & 10 & 9 & 6 & 5 & 3. \end{array}$$

The first implementation compares hyperplane-coordinate orbit prefixes. A second implementation shares the first-independent-basis normalization and prefix-search architecture, but independently implements the finite-field operations and compares Fano line-position prefixes directly. Both return exactly these four sequences. This is a cross-check of the invariant implementation, not a fully independent search design.

In the first two types, the first three nonzero vectors in every three-dimensional subspace form a two-dimensional subspace. In the last two types, the corresponding statement holds for the last three nonzero vectors. The first two types induce distinct ordered Fano types. $\square$

The finite lemma can also be viewed recursively: in dimension four, compatibility is equivalent to requiring all hyperplanes to have one common ordered Fano type. Every binary order in dimension three with zero extremal is itself compatible, so no lower-dimensional condition is omitted.

4 From a local line to a global flag

Lemma 4.1 (Initial flag). *Let $n \geq 3$, and let*

$$p_1 \prec p_2 \prec \cdots \prec p_{2^n-1}$$

order the nonzero vectors of $V = \mathbb{F}_2^n$. Suppose that in every three-dimensional subspace, the first three nonzero vectors form a two-dimensional subspace. Then, for every $1 \leq r \leq n$, the first $2^r - 1$ nonzero vectors form an r -dimensional subspace.

Proof. The assertion is immediate for $r = 1$. The vectors p_1, p_2 are distinct and therefore independent. For any $x \notin \langle p_1, p_2 \rangle$, the first two vectors in $\langle p_1, p_2, x \rangle$ are p_1, p_2 . Its third vector must complete their two-dimensional subspace, so it is $p_1 + p_2$. Hence the first three global vectors form a two-space.

Assume the first $2^r - 1$ vectors form an r -space E_r . Let p be the next vector and set $E_{r+1} = E_r + \langle p \rangle$. For $y \notin E_{r+1}$ and $0 \neq u \in E_r$, consider

$$S = \langle u, p, y \rangle.$$

Because $y \notin E_r + \langle p \rangle$, one has $S \cap E_r = \langle u \rangle$. Thus u is the first nonzero vector of S , and p is the second: every other point outside E_r occurs no earlier than the globally first such point p . The local hypothesis forces the third point of S to be $u + p$. Consequently $u + p \prec y$ for every nonzero $u \in E_r$ and every $y \notin E_{r+1}$. Together with $p \prec y$, the whole coset $p + E_r$ precedes every vector outside E_{r+1} . Hence the first $2^{r+1} - 1$ vectors are exactly $E_{r+1} \setminus \{0\}$. Induction completes the proof. $\square$

5 Unique lifting and the classification

Lemma 5.1 (Unique lift). *Fix an ordered Fano type T whose first three points form a line. For every $n \geq 3$, there is at most one $\mathrm{GL}(n, 2)$ -orbit of orders on $\mathbb{F}_2^n \setminus \{0\}$ satisfying both:*

1. *every three-dimensional subspace induces type T ;*
2. *for every r , the first $2^r - 1$ vectors form an r -subspace.*

Proof. We induct on n . Suppose two candidate orders have initial hyperplanes E, E' . By the induction hypothesis, a linear isomorphism identifies their induced orders on these hyperplanes. Let p, p' be the first vectors outside E, E' . Extend the hyperplane isomorphism linearly by sending p to p' . We may therefore assume that both candidates agree on E , have the same p , and are defined on the same vector space.

Take two vectors $p + u, p + v$ in the outside coset. Choose a two-dimensional subspace $K \leq E$ containing u and v ; if u, v span a smaller space, extend it to such a K . Put $S = K \oplus \langle p \rangle$. Since every nonzero vector of E precedes p , the first three nonzero vectors of S are precisely the three points of K , in the same order in both candidates, and the fourth is p . Both restrictions have type T . Their order-preserving linear isomorphism therefore fixes two basis vectors of K and fixes p , hence is the identity on S . In particular, the relative order of $p + u$ and $p + v$ is the same in both candidates. This holds for every pair in the outside coset, so the two full orders agree. $\square$

Theorem 5.2. *For every $n \geq 4$, compatible total orders on $\mathbb{F}_2^n$ form exactly eight orbits under $\mathrm{GL}(n, 2)$. Representatives are*

$$L_n, R_n, \rho L_n, \rho R_n, \iota L_n, \iota R_n, \iota \rho L_n, \iota \rho R_n.$$

Consequently, the number of labelled compatible total orders is

$$8|\mathrm{GL}(n, 2)| = 8 \prod_{i=0}^{n-1} (2^n - 2^i).$$

Proof. The one-dimensional subspaces show that zero has the same rank in every two-point subspace. Hence zero is globally least or greatest. Reverse the full order if necessary and assume zero is least.

All four-dimensional subspaces have one common ordered linear type. By Lemma 3.1 this type is one of $L_4, R_4, \rho L_4, \rho R_4$. If it is one of the last two, apply ρ to the global order. Every three-dimensional subspace is contained in a four-dimensional one, so in the resulting order its first three nonzero vectors form a line. Lemma 4.1 supplies a complete initial flag.

The common ordered Fano type is now one of the two distinct types induced by L_4 and R_4 . Lemma 5.1 gives at most one orbit above each type, while Proposition 2.1 supplies L_n and R_n . Undoing ρ gives exactly four orbits with zero least. Full reversal gives four with zero greatest.

Finally, the action of $\text{GL}(n, 2)$ on total orders is free. A linear map preserving a finite total order fixes each vector in its position and is therefore the identity. Every orbit has size $|\text{GL}(n, 2)|$, proving the formula. $\square$

Corollary 5.3. *The numbers in dimensions one, two and three are*

$$2, \quad 2 \cdot 3! = 12, \quad 2 \cdot 7! = 10080,$$

respectively.

Proof. Zero must be extremal. For $n = 1$ there are two orders. For $n = 2$, any ordering of the three nonzero vectors is compatible. The same is true for the seven nonzero vectors when $n = 3$: the only nontrivial proper subspaces are Fano lines, and every bijection between two binary two-spaces that fixes zero is linear. $\square$

6 Reproducibility and scope

The four-dimensional certificate is supplied by `classify_n4.py`, the hyperplane-coordinate prefix search, and `verify_n4_independent.py`, the line-position-prefix verifier. The corresponding JSON result files have SHA256 hashes

32b5a7c748270696372f57e5df890c963a4aef09cf583bd9c5e95cc2a642663a
cb682eb6e88009f7cda3b8c127f7fc323ac30c6a7910a0c0886e6ae328c3753f.

The first result was made deterministic by excluding wall-clock timing from the certificate. Finite recursive searches in dimensions five through seven guided the discovery of the unique-lift lemma, but no claim from those searches is used in the manuscript or included in this certificate.

The theorem answers the BCC30 counting problem over $\mathbb{F}_2$. It does not classify compatible orders for $q > 2$, where zero need not be extremal and each projective point contains more than one nonzero vector. Our literature check covered Cameron’s original notes, the BCC30 restatement, and direct searches using the definition and the expected counting formula. We found no source that states the binary classification. This records the checked boundary and is not a claim that unpublished or unindexed parallel work cannot exist.

AI-assisted research disclosure

Carptopus is the responsible author and takes responsibility for the claims, proofs, source use, and released artifacts. OpenAI Codex was used for AI-assisted literature organization, proof development and stress testing, exact verification, adversarial auditing, and manuscript preparation.

References

- [1] Peter J. Cameron, *Combinatorics of inverse semigroups*, expository notes. <https://webpace.maths.qmul.ac.uk/p.j.cameron/csgnotes/invsemi.pdf>.
- [2] Peter J. Cameron (ed.), *Problems from BCC30*, arXiv:2409.07216v1 (2024), Problem 30.7. <https://arxiv.org/abs/2409.07216>.